Satya Sanjay Chopra

ch24btech11033@iith.ac.in | satyachopra0709@gmail.com | +91 9819026005 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.72
New Bombay City School	Class XII (CBSE)	2024	91.20%
Avalon Heights International School	Class X (ICSE)	2022	96.20%

SCHOLASTIC ACHIEVEMENTS

- Enrolled in B.S in Data Science and Applications at IIT Madras with CGPA of 9.25
- Topper in the following courses in B.S degree : Programming in Python , Statistics , Computational Thinking

WORK EXPERIENCE

SwapR - Hyderabad | April 2025 - Present

CEO and Founder

- Conceptualized and launched a mobile application on the Google Play Store designed to facilitate seamless train seat exchanges across the Indian railway network using a reward system combined with user friendly experience.
- Engineered a comprehensive revenue model, negotiated payment gateway partnerships, setup a business corporation, maintained contracts and mapped the end-to-end user workflow to solve a highly specific consumer pain point.

Diagonal Consulting - Bengaluru | May 2025 - Aug 2025

Analytics and Training Intern

- Mastered business analytics by building data pipelines and designing interactive dashboards using Qlik Sense and Tableau.
- Leveraged SQL and Python to transform complex datasets, ensuring high data integrity for actionable business intelligence.

Bhaichung Bhutia Football Schools (BBFS) - Mumbai | May 2025 - June 2025

Research and Operations Intern

- Conducted field level research and operational analysis for one of India's largest football academies.
- Gained comprehensive insights into grassroots sports management, combining operational strategy with a deep understanding of the footballing ecosystem while exploring feasible and unique marketing alternatives.

PROJECTS AND CERTIFICATIONS

Options Volatility Surface Cleaner & Calibrator

- Engineered a fully automated data processing pipeline using non-linear optimization algorithms to systematically detect and eliminate market arbitrage anomalies and effectively mapping smooth 3D SABR volatility models.
- Architected the computational framework to generate highly accurate, strictly arbitrage-free volatility pricing surfaces for complex financial derivatives, empowering quantitative analysts with reliable market data.

Modern Portfolio Theory & Efficient Frontier Optimizer

- Developed an asset allocation platform to isolate risk-optimized investment structures and maximize portfolio returns.
- Implemented matrix algebra and vectorized Monte Carlo simulations to accurately model multi-asset covariance.

Quantitative Algorithmic Trading & Backtesting Engine

- Programmed an end-to-end historical execution simulator designed to rigorously backtest and optimize momentum based quantitative trading strategies against volatile, real-world market conditions and historical price data.
- Engineered the simulation framework to dynamically account for broker transaction commissions, bid ask spreads, and realistic market slippage friction, ensuring accurate and high-fidelity strategy validation.

Fixed-Income Yield Curve & Risk Simulator

- Created a scalable financial risk simulation engine engineered to evaluate multi-asset portfolio duration sensitivities, evaluate convexity, and proactively manage fixed-income asset exposures across diverse market conditions.
- Built the mathematical framework to dynamically model shifting yield curves under various macroeconomic interest rate shocks and stress testing scenarios, providing portfolio managers with clear, quantifiable risk assessment metrics.

Football Forwards Performance Analysis

- Analyzed key performance metrics of football forwards to identify underlying performance drivers.
- Utilized statistical modeling and data visualization techniques to translate raw match data into actionable insights.

Certifications : Quantium - Data Analytics, **British Airways** - Data Science, **Tata Consultancy Services** - GenAI Data Analytics

ACCOMPLISHMENTS AND LEADERSHIP

- **Advanced to Second Round of Goldman Sachs India Hackathon:** Progressed among top competitors by engineering an algorithmically robust, time-optimized quantitative solution under strict deadlines. Demonstrates high-level quantitative problem solving and the ability to build financial technology to industry standards.

- **Advanced to Round 2 in EY Technathon:** Selected for the second round by developing an innovative, tech-driven prototype to solve complex enterprise business challenges. Highlights a strong aptitude for rapid innovation, analytical structuring, and translating technical concepts into viable commercial applications.

- Participated in prestigious case competitions: **HP Power Lab 2.0, IIM Calcutta Consulting Consortium, IIT Kharagpur CaseForge, and IIT Guwahati ProduScope :** Developed comprehensive go-to-market strategies and operational models in top-tier national challenges, honing cross-functional business acumen and high-pressure problem-solving.

- Appointed as **Head of Sponsorship** for Elan & nVision, the annual techno-cultural fest of IIT Hyderabad, leading key corporate outreach initiatives and negotiating high-value brand partnerships for the festival.

- Appointed as the **youngest-ever Sponsorship Coordinator** for the premier annual techno-cultural festival of IIT Hyderabad, demonstrating exceptional early-stage leadership, negotiation capabilities, and professional maturity.

- Appointed as **Internship Coordinator** at Office of Career Services, IIT Hyderabad

- **Achieved perfect score of 100 in Mathematics** in both 10th and 12th grade board examinations